Solar Sail Momentum Management Using Predictive Control with Disturbance Torque Estimation

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Motivation, Objectives & Approach

Solar Sail

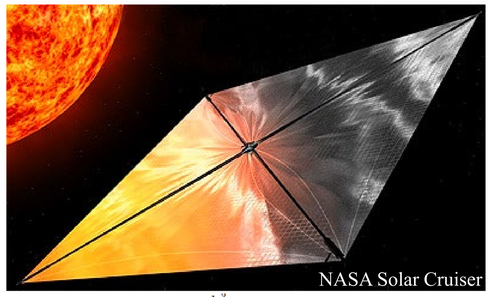
- Leverages solar radiation pressure (SRP) to enable unlimited propulsion, while highly coupled with the sail attitude.

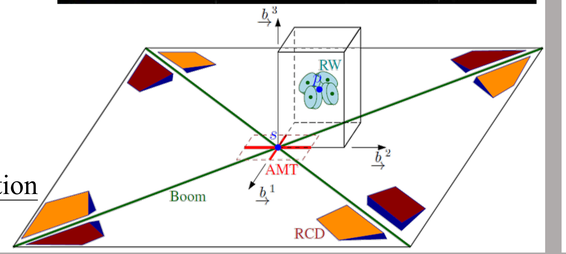
State-of-the-Art Momentum Management & Challenges

- Uses reactive and decoupled controller channels to actuate AMT and RCDs for momentum management.
- Suboptimal performances require innovative controller design.

Model Predictive Control (MPC) Approach

- Incorporates a dynamics model to anticipate angular momentum growth under coupling effects and determine optimal actuation.
- Formulating as a quadratic program (QP) allows efficient, real-time implementation onboard a solar sail.
- Augmenting with Kalman filter disturbance torque estimation provides critical parameters for MPC prediction model.

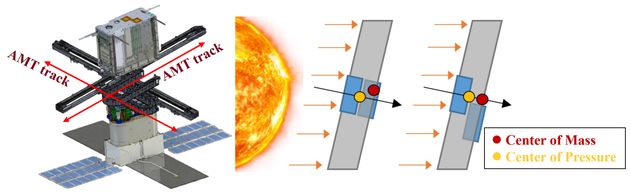




Actuators & Control Inputs

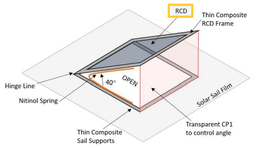
Active Mass Translator (AMT)

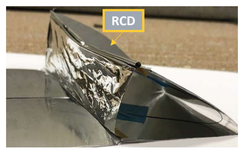
- Continuously changing relative position of CM/CP.



Reflectivity Control Device (RCD)

- Switching on-off commands in CW/CCW directions.





[1] NASA Images

Predictive Control & Disturbance Torque Estimation

System State: $\mathbf{x}_k = \begin{bmatrix} \boldsymbol{\theta}_k^\mathsf{T} & \boldsymbol{\omega}_{b,k}^{ba\mathsf{T}} & \mathbf{h}_{b,k}^{\mathrm{RW}\mathsf{T}} & \mathbf{e}_k^{\mathrm{int}\mathsf{T}} \end{bmatrix}^\mathsf{T}$

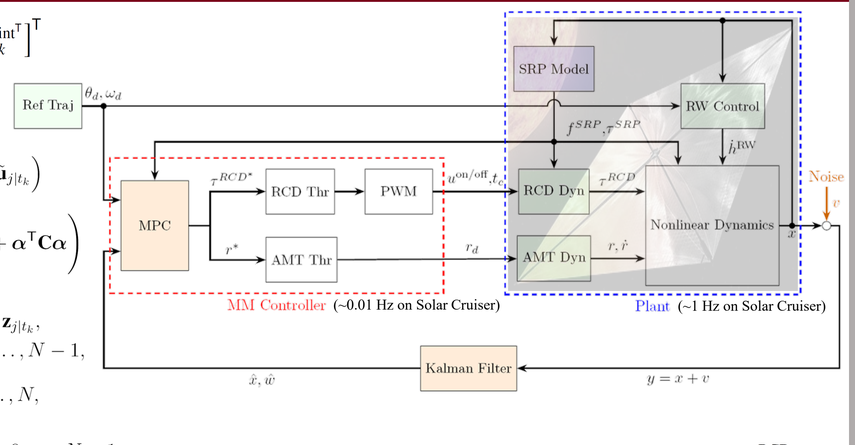
MPC Optimization Problem:

$$\underset{\boldsymbol{\mathcal{X}},\boldsymbol{\mathcal{U}},\boldsymbol{\alpha}\geq\mathbf{0}}{\text{minimize}} \left(\sum_{j=0}^{N-1} \left(\tilde{\mathbf{x}}_{j|t_k}^\mathsf{T} \mathbf{Q} \tilde{\mathbf{x}}_{j|t_k} + \mathbf{u}_{j|t_k}^\mathsf{T} \mathbf{R} \mathbf{u}_{j|t_k} + \tilde{\mathbf{u}}_{j|t_k}^\mathsf{T} \tilde{\mathbf{R}} \tilde{\mathbf{u}}_{j|t_k} \right) + \tilde{\mathbf{x}}_{N|t_k}^\mathsf{T} \mathbf{Q}_N \tilde{\mathbf{x}}_{N|t_k} + \boldsymbol{\alpha}^\mathsf{T} \mathbf{C} \boldsymbol{\alpha} \right)$$

subject to

$$\mathbf{x}_{j+1|t_k} = \mathbf{A}_{j|t_k}\mathbf{x}_{j|t_k} + \mathbf{B}_{u,j|t_k}\mathbf{u}_{j|t_k} + \mathbf{B}_{w,j|t_k}\mathbf{w}_{j|t_k} + \mathbf{z}_{j|t_k},$$
$$\mathbf{x}_{0|t_k} = \mathbf{x}(t_k), \quad \mathbf{u}_{-1|t_k} = \mathbf{u}(t_k), \qquad j = 0, \ldots, N-1,$$
$$\begin{bmatrix} \mathbf{x}_{\min} \\ \mathbf{h}_{\min}^{\mathrm{soft}} - \boldsymbol{\alpha} \end{bmatrix} \leq \begin{bmatrix} \mathbf{x}_{j|t_k} \\ \mathbf{h}_{j|t_k}^{\mathrm{RW}} \end{bmatrix} \leq \begin{bmatrix} \mathbf{x}_{\max} \\ \mathbf{h}_{\max}^{\mathrm{soft}} + \boldsymbol{\alpha} \end{bmatrix}, \quad j = 0, \ldots, N,$$
$$\begin{bmatrix} \mathbf{u}_{\min} \\ \dot{\mathbf{u}}_{\min} \end{bmatrix} \leq \begin{bmatrix} \mathbf{u}_{j|t_k} \\ (\mathbf{u}_{j|t_k} - \mathbf{u}_{j-1|t_k})/\Delta t \end{bmatrix} \leq \begin{bmatrix} \mathbf{u}_{\max} \\ \dot{\mathbf{u}}_{\max} \end{bmatrix}, \quad j = 0, \ldots, N-1.$$

Solve for $\boldsymbol{\mathcal{X}} = \{\mathbf{x}_{0|t_k}, \ldots, \mathbf{x}_{N|t_k}\}, \boldsymbol{\mathcal{U}} = \{\mathbf{u}_{-1|t_k}, \mathbf{u}_{0|t_k}, \ldots, \mathbf{u}_{N-1|t_k}\}, \boldsymbol{\alpha}$, and only apply $\mathbf{u}_{0|t_k}$ at timestep k.

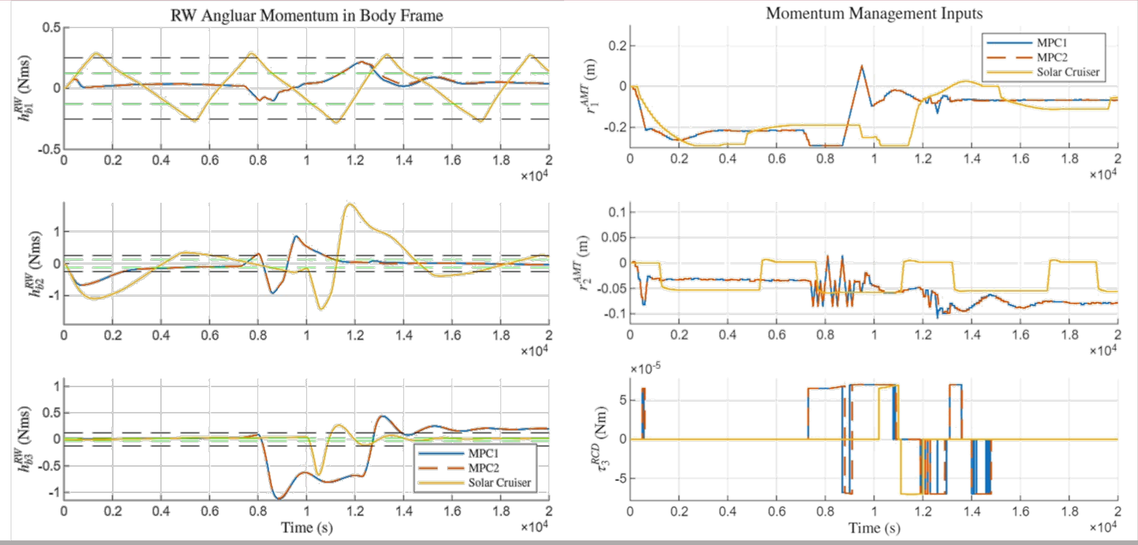


RCD Quantization (PWM): $t_c = \Delta t \cdot \dfrac{u_{k,\mathrm{mpc}}^{\mathrm{RCD}}}{u_{\mathrm{on}}^{\mathrm{RCD}}}$

Simulation Results & State-of-the-Art Comparison

Slew maneuver at $t = 8000$ s.

- **State-of-the-art method** can only perform maneuvers up to $\boldsymbol{\theta}_d = [0°, 10°, 1°]^\mathrm{T}$ without RW saturation.
- **Proposed MPC framework** survives larger maneuvers of $\boldsymbol{\theta}_d = [0°, 10°, 15°]^\mathrm{T}$ and even further.
- Flexible to fit different goals:
  - MPC1 uses Quantization for shorter RCD "on" time;
  - MPC2 includes RCD actuation cost with full-timestep actuation for fewer on-off cycles.



Key Take Aways

- Larger slew maneuver envelope is enabled by MPC-based momentum management over state-of-the-art method.
- Flexible tuning and objective modification can tailor MPC for different actuator limits.
- Collaborating with NASA to develop MPC-based momentum management software for future solar sails.

NASA Partner

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[2] Shen and Caverly, *Acta Astronautica*, 2026.
[3] Shen and Caverly, *Advances in Space Research*, 2026.